

QUANT MOCK PACK (100 Qs)

NN-01 Quantitative Aptitude | 45 minutes | key with trap notes at end

QUESTIONS (100) - tick your answer, 45 minutes

Q1. Which of these numbers is irrational?

- a) $22/7$ b) 0.4545... c) $\sqrt{2}$ d) -3

Q2. Which of these is a prime number?

- a) 1 b) 51 c) 97 d) 91

Q3. odd + odd + even = ?

- a) Odd b) Even c) Depends d) Zero

Q4. In 5708, the place value of 7 is:

- a) 7 b) 70 c) 700 d) 7000

Q5. How many factors does 18 have?

- a) 5 b) 6 c) 4 d) 8

Q6. 100 divided by 7 gives quotient and remainder:

- a) q14 r2 b) q13 r9 c) q14 r0 d) q15 r2

Q7. Which number is divisible by 3?

- a) 7732 b) 9453 c) 7321 d) 1112

Q8. Which number is divisible by 4?

- a) 5678 b) 3124 c) 1235 d) 5671

Q9. Which number is divisible by 6?

- a) 3471 b) 4572 c) 1235 d) 7777

Q10. Which number is divisible by 8?

- a) 5678 b) 9168 c) 1234 d) 7777

Q11. Which number is divisible by 9?

- a) 12345 b) 33660 c) 77777 d) 10101

Q12. Which number is divisible by 11?

- a) 2728 b) 1234 c) 5678 d) 9091

Q13. In $2*5$ (divisible by 11), the digit * is:

- a) 2 b) 4 c) 6 d) 8

Q14. Least to add to 1000 to make it divisible by 9:

- a) 6 b) 7 c) 8 d) 9

Q15. HCF of 36 and 60 is:

- a) 6 b) 12 c) 18 d) 4

Q16. $H = 8$, $L = 60$, one number 20, the other is:

- a) 22 b) 24 c) 26 d) 30

Q17. $3/5 + 1/10 = ?$

- a) $4/10$ b) $7/10$ c) $1/2$ d) $4/5$

Q18. $0.6 \times 0.05 = ?$

- a) 0.3 b) 0.03 c) 0.003 d) 3

Q19. 62.5% as a fraction is:

a) $\frac{5}{8}$ b) $\frac{3}{5}$ c) $\frac{2}{3}$ d) $\frac{7}{8}$

Q20. Which is bigger: $\frac{4}{7}$ or $\frac{5}{9}$?

a) $\frac{4}{7}$ b) $\frac{5}{9}$ c) Equal d) Cannot say

Q21. Is $\frac{9}{40}$ terminating or recurring?

a) Terminating b) Recurring c) Neither d) Both

Q22. $20 - 4 \times 3 + 2 = ?$

a) 10 b) 12 c) 54 d) 6

Q23. $5 - [2 + 3 \times 2] = ?$

a) -3 b) 11 c) 1 d) -1

Q24. $(\frac{5}{8}) \div (\frac{15}{16}) = ?$

a) $\frac{2}{3}$ b) $\frac{3}{2}$ c) $\frac{5}{6}$ d) $\frac{1}{2}$

Q25. $\sqrt{169} + \text{cube-root}(64) = ?$

a) 15 b) 17 c) 21 d) 13

Q26. Simplify $\sqrt{72}$:

a) $6\sqrt{2}$ b) $8\sqrt{2}$ c) $36\sqrt{2}$ d) $9\sqrt{2}$

Q27. Divide 45 in 2:3:5. First share?

a) 9 b) 13.5 c) 22.5 d) 15

Q28. If $\frac{x}{5} = \frac{12}{20}$, then $x = ?$

a) 2 b) 3 c) 4 d) 5

Q29. 3 pens cost 45. 7 pens cost?

a) 95 b) 100 c) 105 d) 110

Q30. 8 pipes fill in 6 h. 12 pipes take?

a) 2 h b) 3 h c) 4 h d) 5 h

Q31. Divide 600 in 1:2:3. Middle share?

a) 100 b) 150 c) 200 d) 300

Q32. Compound ratio $(3:4)(5:7) = ?$

a) 8:11 b) 15:28 c) 3:7 d) 5:4

Q33. 20% of 350 = ?

a) 60 b) 70 c) 75 d) 80

Q34. 45% rises to 54%. Rise in points?

a) +9 points b) +9% c) +20 points d) +20%

Q35. Population 80,000 falls 15%. New population?

a) 68000 b) 65000 c) 70000 d) 72000

Q36. Average of 8 numbers is 15; 7 of them sum to 96. Missing?

a) 20 b) 24 c) 26 d) 30

Q37. Combined average of 40@60 and 60@90?

a) 72 b) 75 c) 78 d) 80

Q38. Average speed for 60 + 90 over equal distance?

a) 72 b) 74 c) 75 d) 70

Q39. CP 300, SP 375. Profit%?

a) 20 b) 25 c) 30 d) 35

Q40. MP 2000, SP 1700. Discount%?

a) 10 b) 12 c) 15 d) 20

Q41. Net result of -15% then -10%?

a) -25% b) -23.5% c) -25.5% d) -20%

Q42. After -20% the price is 320. Original?

a) 380 b) 400 c) 384 d) 360

Q43. Sold 950 g for 1000 g. Gain%?

a) 5% b) 5.26% c) 6% d) 4.5%

Q44. SI on 8000 at 5% for 4 years?

a) 1200 b) 1600 c) 2000 d) 1500

Q45. SI 600 at 5% for 4 years. Principal?

a) 2500 b) 3000 c) 3500 d) 2000

Q46. Money triples in 20 years. Rate%?

a) 8 b) 10 c) 12 d) 15

Q47. CI on 5000 at 8% for 2 years (interest)?

a) 800 b) 832 c) 900 d) 1000

Q48. CI-SI difference: 4000 at 5% for 2 years?

a) 8 b) 10 c) 12 d) 20

Q49. 6000x9 : 9000x6, profit 13000. Shares?

a) 6500 each b) 6000/7000 c) 7000/6000 d) 5200/7800

Q50. Profit 13000, A salary 1000, ratio 1:1. B total?

a) 6000 b) 6500 c) 7000 d) 7500

Q51. Average of Rs.30x2kg + Rs.50x3kg?

a) 40 b) 42 c) 44 d) 46

Q52. Mix 10 and 25 for mean 20. Ratio?

a) 1:2 b) 2:1 c) 1:1 d) 3:2

Q53. 80 L, 20 L out twice. Milk left?

a) 40 L b) 45 L c) 50 L d) 55 L

Q54. Father:son 5:1, difference 24. Father?

a) 24 b) 28 c) 30 d) 36

Q55. A 10 days + B 15 days together?

a) 5 d b) 6 d c) 7 d d) 8 d

Q56. A+B 10 days, wages 1:2 of 6000. A gets?

a) 1500 b) 2000 c) 2500 d) 3000

Q57. 20 men 12 days = 30 men ?

a) 6 d b) 8 d c) 9 d d) 10 d

Q58. A 5d, B 10d, alternate B-first. Total?

a) 6 d b) 7 d c) 8 d d) 9 d

Q59. A 6d, B 12d, wages 9000. A share?

a) 4500 b) 5000 c) 6000 d) 6500

Q60. Fill 4h + empty 12h. Net fill?

a) 4 h b) 5 h c) 6 h d) 8 h

Q61. Fills in 6h, with leak in 8h. Leak alone?

a) 20 h b) 22 h c) 24 h d) 26 h

Q62. 54 km/h for 40 min. Distance?

a) 30 km b) 32 km c) 36 km d) 40 km

Q63. 15 m/s in km/h?

a) 45 b) 50 c) 54 d) 60

Q64. A60 B90, A 120 km behind. Catch-up time?

a) 3 h b) 4 h c) 5 h d) 6 h

Q65. 30 km@1h + 90 km@2h. Average?

a) 60 b) 65 c) 70 d) 75

Q66. 180 m train at 54 km/h crosses pole in?

a) 10 s b) 12 s c) 14 s d) 15 s

Q67. 200+300 m trains, 90 vs 54 opposite. Time?

a) 10 s b) 12 s c) 12.5 s d) 15 s

Q68. Down 18, Up 10. Boat speed?

a) 12 b) 13 c) 14 d) 15

Q69. Down 18, Up 10. Stream speed?

a) 2 b) 3 c) 4 d) 5

Q70. 72 km, Down 18 Up 12. Round-trip time?

a) 8 h b) 9 h c) 10 h d) 12 h

Q71. Clock angle at 6:00?

a) 90 deg b) 120 deg c) 180 deg d) 0 deg

Q72. Clock angle at 8:24?

a) 90 deg b) 100 deg c) 108 deg d) 120 deg

Q73. Hands coincide how many times per 12 h?

a) 10 b) 11 c) 12 d) 13

Q74. Which is a leap year?

a) 1800 b) 1900 c) 2000 d) 2100

Q75. 2023 calendar repeats in?

a) 2028 b) 2034 c) 2029 d) 2030

Q76. $5x - 11 = 24$. x ?

a) 5 b) 6 c) 7 d) 8

Q77. $2x+3y = 17$, $x-y = 1$. x ?

a) 2 b) 3 c) 4 d) 5

Q78. $98 \times 102 = ?$

a) 9996 b) 10000 c) 9604 d) 10004

Q79. Thrice a number less 4 is 20. Number?

a) 6 b) 7 c) 8 d) 9

Q80. Square perimeter 28. Area?

a) 42 b) 49 c) 56 d) 64

Q81. 15 x 8 rectangle. Area?

a) 100 b) 110 c) 120 d) 130

Q82. Triangle 5, 12, 13. Area?

a) 24 b) 28 c) 30 d) 36

Q83. Equilateral side 4. Area?

a) $4\sqrt{3}$ b) $8\sqrt{3}$ c) $6\sqrt{3}$ d) 16

Q84. Circle r14 (22/7). Area?

a) 616 b) 484 c) 88 d) 44

Q85. Cube side 6. Volume?

a) 216 b) 180 c) 150 d) 200

Q86. Cuboid 4x3x5. Total surface?

a) 60 b) 74 c) 94 d) 100

Q87. Cylinder r14 h20 (22/7). Volume?

a) 12320 b) 12000 c) 10000 d) 15000

Q88. Cone r9 h12 (3.14). Volume?

a) 900 b) 1017 c) 1100 d) 1200

Q89. Sphere r10.5 (22/7). Volume?

a) 4500 b) 4851 c) 5000 d) 5200

Q90. Hemisphere r14. Volume?

a) 5745 b) 5000 c) 6000 d) 4500

Q91. Tank 3x2x1.5 m. Litres?

a) 6000 b) 8000 c) 9000 d) 10000

Q92. Cube 9^3 into 3^3 cubes. Count?

a) 9 b) 18 c) 27 d) 36

Q93. Cube diagonal, side 5?

a) $5\sqrt{2}$ b) $5\sqrt{3}$ c) 25 d) 15

Q94. Cylinder CSA r7 h10?

a) 220 b) 330 c) 440 d) 550

Q95. 30% sector of total 5400?

a) 1500 b) 1620 c) 1800 d) 2000

Q96. Ratio 72 deg : 108 deg?

a) 1:2 b) 2:3 c) 3:4 d) 4:5

Q97. Cone slant height r6 h8?

a) 8 b) 10 c) 12 d) 14

Q98. Hemisphere TSA r7?

a) 308 b) 462 c) 616 d) 231

Q99. Sphere r21 (22/7). Surface?

a) 5544 b) 5000 c) 6000 d) 4500

Q100. 1 cubic metre = ? litres

a) 100 b) 1000 c) 10000 d) 100000

ANSWER KEY (answers + trap notes)

Q1: c - 22/7 only approximates pi; sqrt(2) truly irrational.

Q2: c - 1 neither; $51 = 3 \times 17$; $91 = 7 \times 13$.

Q3: b - odd+odd = even, +even stays even.

Q4: c - 7 in hundreds place = 700; face value 7.

Q5: b - 1,2,3,6,9,18 = six factors.

Q6: a - $6 \times 14 = 98$, remainder 2; remainder < divisor.

Q7: b - Digit sum 21 divides by 3.

Q8: b - Last two 24 divisible by 4.

Q9: b - Even + digit sum 18: both rules for 6.

Q10: b - Last three $168/8 = 21$.

Q11: b - Digit sum 18 divides by 9.

Q12: a - $(2+2)-(7+8) = -11$, multiple of 11.

Q13: b - Present sum 7; needs 4 or 15; digit 4.

Q14: c - Next multiple 1008; add 8.

Q15: b - Division ladder ends at 12.

Q16: b - $b = 8 \times 60/20 = 24$ ($a \times b = H \times L$).

Q17: b - Common denominator 10: $6+1 = 7/10$.

Q18: b - Three decimal places: 0.030.

Q19: a - $62.5\% = 5/8$; memorize percent bank.

Q20: a - Cross: 36 vs 35, 4/7 wins.

Q21: a - $40 = 8 \times 5$, only 2s and 5s.

Q22: a - Multiply first: $20-12+2 = 10$.

Q23: a - Inner first: $5-8 = -3$.

Q24: a - Flip second: $5/8 \times 16/15 = 2/3$.

Q25: b - $13 + 4 = 17$.

Q26: a - $72 = 36 \times 2$, root 36 = 6.

Q27: a - Parts 9: $45 \times 2/9 = 9$, 13.5, 22.5.

Q28: b - Cross: $20x = 60$, $x = 3$.

Q29: c - Unitary: $15 \times 7 = 105$.

Q30: c - Inverse: $8 \times 6/12 = 4$ h.

Q31: c - $600/6 = 100$ per part; middle 200.

Q32: b - Multiply: $3 \times 5 : 4 \times 7 = 15:28$.

Q33: b - $350 \times 20/100 = 70$.

Q34: a - +9 points (which is +20%, not +9%).

Q35: a - $80k \times 0.85 = 68k$.

Q36: b - Total $120 - 96 = 24$.

Q37: c - $7800/100 = 78$; never average averages.

Q38: a - $2 \times 60 \times 90/150 = 72$, not 50.

Q39: b - $75/300 = 25\%$ on CP.

Q40: c - $300/2000 = 15\%$ on MP.

Q41: b - $-15-10+1.5 = -23.5$; never add.

Q42: b - Reverse: $320 \times 100/80 = 400$.

Q43: b - $50/950 = 5.26\%$ on actual, not claim.

Q44: b - $8000 \times 5 \times 4/100 = 1600$.

Q45: b - $P = 600 \times 100 / 20 = 3000$.
Q46: b - $R = 200 / 20 = 10\%$.
Q47: b - $5000 \times 1.1664 = 5832$, interest 832.
Q48: b - $4000 \times 25 / 10000 = 10$.
Q49: a - $54000 : 54000 = 1 : 1$, 6500 each.
Q50: c - Rest 12000 split 6000; $B + \text{salary } 1000 = 7000$.
Q51: b - $210 / 5 = 42$; weight the prices.
Q52: a - $(25 - 20) : (20 - 10) = 5 : 10 = 1 : 2$.
Q53: b - $80 \times 0.75^2 = 45$ L; power repeats.
Q54: c - 4 parts $x = 6$: ages 30 and 6.
Q55: b - $150 / 25 = 6$ days; add rates.
Q56: b - Inverse of days 2:1; $A = 2000$.
Q57: b - $20 \times 12 / 30 = 8$ days.
Q58: b - 3 cycles reach 9/10 in 6d; day 7 B finishes.
Q59: c - Inverse 12:6 = 2:1; $A = 6000$.
Q60: c - $1/4 - 1/12 = 1/6$, 6 h; leak negative.
Q61: c - $1/6 - 1/8 = 1/24$, 24 h.
Q62: c - $54 \times 40 / 60 = 36$ km.
Q63: c - $15 \times 18 / 5 = 54$ km/h.
Q64: b - Diff 30; $120 / 30 = 4$ h; same-dir subtract.
Q65: c - $210 / 3 = 70$; total/total-time.
Q66: b - 15 m/s; $180 / 15 = 12$ s.
Q67: c - $144 \text{ km/h} = 40 \text{ m/s}$; $500 / 40 = 12.5$ s.
Q68: c - $b = (18 + 10) / 2 = 14$.
Q69: c - $s = (18 - 10) / 2 = 4$.
Q70: c - $4 + 6 = 10$ h; each leg separate.
Q71: c - Opposite directions: 180 deg.
Q72: c - $|240 - 132| = 108$ deg; both hands.
Q73: b - 11 times, not 12.
Q74: c - Century only if /400.
Q75: b - Ordinary +11: 2034.
Q76: c - Isolate: $5x = 35$.
Q77: c - $x = y + 1$ gives $5y = 15$; $x = 4$.
Q78: a - $(100 - 2)(100 + 2) = 10000 - 4$.
Q79: c - $3x = 24$.
Q80: b - Side 7; 49.
Q81: c - 120 area; 46 perimeter.
Q82: c - Half of 60 (5-12-13 triple).
Q83: a - $\sqrt{3} / 4 \times 16 = 4 \sqrt{3}$.
Q84: a - $22/7 \times 196 = 616$.
Q85: a - 216 volume; surface also 216.
Q86: c - $2(20 + 15 + 12) = 94$.
Q87: a - $22/7 \times 196 \times 20 = 12320$.
Q88: b - One-third $\times 81 \times 12 \times 3.14 = 1017$.
Q89: b - $4/3 \times 22/7 \times 1157.6 = 4851$.

Q90: a - $\frac{2}{3} \times \frac{22}{7} \times 2744 = 5745$.

Q91: c - 9 cubic m = 9000 L.

Q92: c - $\frac{729}{27} = 27$; volume conserved.

Q93: b - Space diagonal a $\sqrt{3}$.

Q94: c - $2 \times \frac{22}{7} \times 7 \times 10 = 440$.

Q95: b - $5400 \times 0.3 = 1620$.

Q96: b - Divide 36: 2:3.

Q97: b - $\sqrt{36+64} = 10$; slant via Pythagoras.

Q98: b - $3 \times \frac{22}{7} \times 49 = 462$; base included.

Q99: a - $4 \times \frac{22}{7} \times 441 = 5544$.

Q100: b - 1000 L; convert before comparing.